
DBSolo Setup Runbook for QBO

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Overview

This WIKI walks us through the steps for setting up DBSolo and Data Comparison Reports for QBO.

Prerequisites

dos2unix

html2text

AWS S3 bucket

Installation

Ref to [Data Comparison between Oracle and Postgres Using DB Solo#DBSoloInstallation](#) for detailed installation steps

Data Comparison Reports Setup

~~For QBO, DBSolo home directory is /u01/ogg/scripts/dbsolo5. I will be using the same for the commands in this wiki.~~

You should see the following files under /u01/ogg/scripts/dbsolo5-

Name
 01_dbsolo_create_setup_templates.sh
 02a_dbsolo_where_clause_custom_days.sh
 02b_dbsolo_where_clause_custom_days_oos.sh
 02c_dbsolo_where_clause_custom_date.sh
 02d_dbsolo_where_clause_custom_date_oos.sh
 03a_dbsolo_where_clause_generic_days.sh
 03b_dbsolo_where_clause_generic_days_oos.sh
 03c_dbsolo_where_clause_generic_date.sh
 03d_dbsolo_where_clause_generic_date_oos.sh
 03e_dbsolo_where_clause_generic_date_jsonb.sh
 03f_dbsolo_where_clause_generic_date_jsonb_oos.sh
 04_dbsolo_where_clause_empty.sh
▼  etc
 generate_03a_dbsolo_where_clause_generic_days.sql
 generate_03b_dbsolo_where_clause_generic_days_oos.sql
 generate_03c_dbsolo_where_clause_generic_date.sql
 generate_03d_dbsolo_where_clause_generic_date_oos.sql
 generate_03e_dbsolo_where_clause_generic_jsonb.sql
 generate_03f_dbsolo_where_clause_generic_jsonb_oos.sql
 generate_qbo_commandline_all.sql
 generate_qbo_commandline_jsonb.sql
 jsonb_start_end.txt
 post_reboot_setup.sh
 qbo_commandline_all.sh
 qbo_commandline_oos.sh
 qbo_commandline.sh
 qbo_schema_tables_1.xml
 qbo_tables.txt
 qbodata_schema_tables_1.xml
 qbodata_tables_jsonb.txt
 qbodata_tables.txt
 README.md

```
/u01/ogg/scripts/dbsolo5/01_dbsolo_create_setup_templates.sh
```

Notes: This command takes care of the following -

- Takes backups of the xml projects **qbo_schema_tables_1.xml** and **qbodata_schema_tables_1.xml**
 - **qbodata_schema_tables_1.xml.bak**
 - **qbo_schema_tables_1.xml.bak**
- Merges both the xml projects for QBO & QBO_DATA schemas and, creates five Report Templates
 - **qbo_qbodata_schemas_tables_template_1.xml**
 - **qbo_qbodata_schemas_tables_template_2.xml**
 - **qbo_qbodata_schemas_tables_template_3.xml**
 - **oos_qbo_qbodata_schemas_tables_template_1.xml**
 - **oos_qbo_qbodata_schemas_tables_template_2.xml**
- Replaces the below hard coded values in both the Report Templates with "variables". We get the hardcoded values in the xml projects when it is generated/refreshed from the database. **Remember if multiple users are generating/refreshing the xml projects, then each user may have different settings on their local DBsolo. In this case DO ENSURE we are replacing the right hardcoded values with this script 01_dbsolo_create_setup_templates.sh. Or everyone uses the same connection.xml. (Per User license is a MUST)**
 - a) source name
 - b) schema
 - c) destination
 - d) datacomp_temp_dir (*Needed if datacomp_filebased value="true"*)
 - e) datacomp_html_file
 - f) datacomp_sync_directoryname

3. Execute the below command (*If you are doing the setup for the first time then please see the Notes section first*)

```
/u01/ogg/scripts/dbsolo5/02a_dbsolo_where_clause_custom_days.sh
```

```
/u01/ogg/scripts/dbsolo5/02b_dbsolo_where_clause_custom_days_oos.sh
```

2. Uses below condition/WHERE CLAUSE for the same 11 tables-

- a. For utc column
 - i. **Oracle**- "*sys_extract_utc(systimestamp)-v_days and sys_extract_utc(systimestamp) -v_hours/24*" ;
 - ii. **Postgres**- "*(CURRENT_TIMESTAMP at time zone 'utc' - INTERVAL 'v_days DAY') and (CURRENT_TIMESTAMP at time zone 'utc' - INTERVAL 'v_hours Hour') "*"
- b. For date column
 - i. **Oracle** - "*between sysdate- v_days and sysdate -v_hours/24*" ;
 - ii. **Postgres** - "*CURRENT_TIMESTAMP - INTERVAL 'v_days DAY' and CURRENT_TIMESTAMP - INTERVAL 'v_hours Hour' "*"
3. Sets "Included" value to *false* for these 11 tables in **qbo_qbodata_schemas_tables_template_1.xml**. When the template is generated, "Included" value has to be *true* else no column list will be generated.
4. Second script does all the above with one additional step for Report template **oos_qbo_qbodata_schemas_tables_template_1.xml**. It adds OOS company_id in the WHERE CLAUSE

4. Execute the below command (*If you are doing the setup for the first time then please see the Notes section first*)

```
/u01/ogg/scripts/dbsolo5/02c_dbsolo_where_clause_custom_date.sh

/u01/ogg/scripts/dbsolo5/02d_dbsolo_where_clause_custom_date_oos.sh
```

Notes: This command takes care of the following in the Report template **qbo_qbodata_schemas_tables_template_2.xml**

1. Updates the WHERE Clause for 11 tables where we are using a subquery or an uncommon date column
2. Uses below condition/WHERE CLAUSE -
 - a. For utc column
 - i. **Oracle** - "*CAST(to_timestamp('v_sdate','yyyy-mm-dd') AT TIME ZONE 'UTC' AS DATE) and CAST(to_timestamp('v_edate','yyyy-mm-dd') AT TIME ZONE 'UTC' AS DATE)*";
 - ii. **Postgres** - "*('v_sdate' at time zone 'utc') and 'v_edate' at time zone 'utc')*"
 - b. For date column
 - i. **Oracle** - "*to_date('v_sdate','yyyy-mm-dd') and to_date('v_edate','yyyy-mm-dd')*" ;
 - ii. **Postgres** - "*'v_sdate' and 'v_edate' "*"
3. Sets "Included" value to *false* for these 11 tables in **qbo_qbodata_schemas_tables_template_2.xml**
4. Second script does all the above with one additional step for Report template **oos_qbo_qbodata_schemas_tables_template_2.xml**. It adds OOS company_id in the WHERE CLAUSE

5. Execute the below command (*If you are doing the setup for the first time then please see the Notes section first*)

```
/u01/ogg/scripts/dbsolo5/03b_dbsolo_where_clause_generic_days_oos.sh
```

Notes: This command takes care of the following in the Report template **qbo_qbodata_schemas_tables_template_1.xml**

1. Updates the WHERE Clause for all the tables where we are using *last_modify_date/last_modified_utc* column
2. Uses condition below condition/WHERE CLAUSE -
 - a. For LAST_MODIFIED_UTC
 - i. **Oracle** - "*sys_extract_utc(systimestamp)-v_days and sys_extract_utc(systimestamp) -v_hours/24*";
 - ii. **Postgres** - "*(CURRENT_TIMESTAMP at time zone 'utc' - INTERVAL 'v_days DAY') and (CURRENT_TIMESTAMP at time zone 'utc' - INTERVAL 'v_hours Hour') "*"
 - b. For LAST_MODIFY_DATE
 - i. **Oracle** - "*between sysdate- v_days and sysdate -v_hours/24*";
 - ii. **Postgres** - "*CURRENT_TIMESTAMP - INTERVAL 'v_days DAY' and CURRENT_TIMESTAMP - INTERVAL 'v_hours Hour'*"
3. Sets "Included" value to *false* for these tables in **qbo_qbodata_schemas_tables_template_1.xml**. When the template is generated, "Included" value has to be *true* else no column list will be generated.
4. Second script does all the above with one additional step for Report template **oos_qbo_qbodata_schemas_tables_template_1.xml**. It adds OOS company_id in the WHERE CLAUSE

6. Execute the below command (*If you are doing the setup for the first time then please see the Notes section first*)

```
/u01/ogg/scripts/dbsolo5/03c_dbsolo_where_clause_generic_date.sh
```

```
/u01/ogg/scripts/dbsolo5/03d_dbsolo_where_clause_generic_date_oos.sh
```

Notes: This command takes care of the following in the Report template **qbo_qbodata_schemas_tables_template_2.xml**

1. Updates the WHERE Clause for all the tables where we are using *last_modify_date/last_modified_utc* column
2. Uses condition below condition/WHERE CLAUSE
 - a. For LAST_MODIFIED_UTC
 - i. **Oracle** - "*CAST(to_timestamp('v_sdate','yyyy-mm-dd') AT TIME ZONE 'UTC' AS DATE) and CAST(to_timestamp('v_edeate','yyyy-mm-dd') AT TIME ZONE 'UTC' AS DATE)*";
 - ii. **Postgres** - "*('v_sdate' at time zone 'utc') and 'v_edeate' at time zone 'utc')*"
 - b. For LAST_MODIFY_DATE
 - i. **Oracle** - "*to_date('v_sdate','yyyy-mm-dd') and to_date('v_edeate','yyyy-mm-dd')*";
 - ii. **Postgres** - "*'v_sdate' and 'v_edeate'*"
3. Sets "Included" value to *false* for these tables in **qbo_qbodata_schemas_tables_template_2.xml**
4. Second script does all the above with one additional step for Report template **oos_qbo_qbodata_schemas_tables_template_2.xml**. It adds OOS company_id in the WHERE CLAUSE

7. Execute the below command ([If you are doing the setup for the first time then please see the Notes section first](#))

```
/u01/ogg/scripts/dbsolo5/03e_dbsolo_where_clause_generic_date_jsonb.sh

/u01/ogg/scripts/dbsolo5/03f_dbsolo_where_clause_generic_date_jsonb_oos.sh
```

Notes: This command takes care of the following in the Report template **qbo_qbodata_schemas_tables_template_3.xml**

1. Updates the WHERE Clause for all the 17 tables with JSONB column where we are using *last_modify_date/last_modified_utc* column
2. Uses condition below condition/WHERE CLAUSE
 - a. For LAST_MODIFIED_UTC
 - i. **Oracle** - "*CAST(to_timestamp('v_sdate','yyyy-mm-dd') AT TIME ZONE 'UTC' AS DATE) and CAST(to_timestamp('v_edate','yyyy-mm-dd') AT TIME ZONE 'UTC' AS DATE)*";
 - ii. **Postgres** - "*('v_sdate' at time zone 'utc') and 'v_edate' at time zone 'utc')*"
 - b. For LAST_MODIFY_DATE
 - i. **Oracle** - "*to_date('v_sdate','yyyy-mm-dd') and to_date('v_edate','yyyy-mm-dd')*";
 - ii. **Postgres** - "*'v_sdate' and 'v_edate'*"
3. Sets "Included" value to *false* for these tables in **qbo_qbodata_schemas_tables_template_3.xml**
4. Reports only the PK column and the JSONB column in the HTML Report
5. Second script does all the above with one additional step for Report template **oos_qbo_qbodata_schemas_tables_template_3.xml**. It adds OOS company_id in the WHERE CLAUSE

8. Execute the below command ([If you are doing the setup for the first time then please see the Notes section first](#))

```
/u01/ogg/scripts/dbsolo5/04_dbsolo_where_clause_empty.sh
```

Notes:

1. There are 29 tables in QBO_DATA schema with no common date column. This command simply sets "Included" value to *false* for these tables in both the Report Templates
2. For most of the QBO schema tables we cannot use date column. Its sets "Included" value to *false* for those QBO Schema .
3. It also sets "Included" value to *false* for the 17 tables in Templates 3 which is only for tables with JSONB column.

10. Verify the memory setting. Change it as per your need in the DBSolo executable file under DBSolo home dir. ([If you are doing the setup for the first time then please see the Notes section first](#))

view /u01/ogg/scripts/dbsolo5/commandLine

change "-Xms128m" "-Xmx512m" if needed to

```
if [ "$has_space_options" = "true" ]; then
$INSTALL4J_JAVA_PREFIX exec "$app_java_home/bin/java" "-Xms512m" "-Xmx102400m" "-Dfile.encoding=UTF-8" "-DdbsoloCom
ARAMS -classpath "$local_classpath" install4j.com.dbsolo.ui.MainApp_commandLine "$@"
return_code=$?
else
$INSTALL4J_JAVA_PREFIX exec "$app_java_home/bin/java" "-Xms512m" "-Xmx102400m" "-Dfile.encoding=UTF-8" "-DdbsoloCom
dbsolo.ui.MainApp_commandLine "$@"
return_code=$?
fi
```

Notes: 1. You can have multiple copies of this file "commandLine" with different memory settings. Each copy can be used to run a report (even in parallel).

2. For automation and customization, we are using a wrapper shell script to call "commandLine". If you want to use a copy with different memory setting then **DO Ensure** the wrapper script is calling the right file. Or you can make a copy of the wrapper script accordingly.

11a. Execute the below command to start Data comparison for a table for a given schema ([If you are doing the setup for the first time then please see the Notes section first](#))

(**qbo_commandline.sh** - wrapper script over DBSolo executable)

```
Usage For Template 1 & 2
./qbo_commandline.sh <clu_name> <src_conn> <trg_conn> <pg_db> <template_xml_file> <tablename> <days for comparison>

Example -

For Template 1 -
nohup ./qbo commandline.sh c4 c4ora c4aur pqbopc04p qbo qbodata schemas tables template 1.xml items 1 10 1 QBO DAT
```


For Template 2 -

```
nohup ./qbo_commandline.sh c4 c4ora c4aur pqbopc04p qbo_qbodata_schemas_tables_template_2.xml items_1 '2021-04-01'
nohup ./qbo_commandline.sh c4 c4ora c4aur pqbopc04p qbo_qbodata_schemas_tables_template_2.xml versions_1 '2021-04-
```

For Template 3 (Only **for** one of the 17 QBO_DATA tables with JSONB column) -

```
nohup ./qbo_commandline.sh c4 c4ora c4aur pqbopc04p qbo_qbodata_schemas_tables_template_3.xml TXHEADERS_1 '2021-04-
```

Notes:

1. Please ensure you are providing correct input values
2. If you have copied the connections.xml from some other environment, please check if all correct for current environment.
3. fyi, connections.xml location - **/home/oracle/dbsolo/**
4. Generated Report will be available locally on the DBSolo ec2 instance , not on S3. (Location - /u01/ogg/scripts/dbsolo5/comparison_reports)
5. No email notification for completion. Email notification only on errors.
6. **This script provides an option to run the report for one table or for all tables. But with ALL option, further automation will not work. Use it only for adhoc purposes.**

11b. Execute the below command to start Data comparison for all the tables in QBO schema or QBO_DATA schema or **BOTH** (**If you are doing the setup for the first time then please see the Notes section first**)

(**qbo_commandline_all.sh** - wrapper script over the script used in step 10a.)

Usage For Template 1 & 2

```
./qbo_commandline_all.sh <clu_name> <src_conn> <trg_conn> <pg_db> <template_xml_file> <days for comparison OR Star
```

Example -

For Template 1 -

```
nohup ./qbo_commandline_all.sh c4 c4ora c4aur pqbopc04p qbo_qbodata_schemas_tables_template_1.xml 10 1 BOTH &
```

```
nohup ./qbo_commandline_all.sh c4 c4ora c4aur pqbopc04p qbo_qbodata_schemas_tables_template_1.xml 10 1 QBO_DATA &
```

```
nohup ./qbo_commandline_all.sh c4 c4ora c4aur pqbopc04p qbo_qbodata_schemas_tables_template_1.xml 10 1 QBO 5 &
```

For Template 2 -

```
nohup ./qbo_commandline_all.sh c4 c4ora c4aur pqbopcv4p qbo_qbodata_schemas_tables_template_2.xml '2021-04-01' 20
```

```
nohup ./qbo_commandline_all.sh c4 c4ora c4aur pqbopc04p qbo_qbodata_schemas_tables_template_2.xml '2021-04-01' 20
```

Usage For Template 3 (Runs only **for** the 17 QBO_DATA tables with JSONB column)

```
./qbo_commandline_all.sh <clu_name> <src_conn> <trg_conn> <pg_db> jsonb_tables_only <End Date (optional)> <Start Date (optional)>
```

Example -

For Template 3 -

```
nohup ./qbo_commandline_all.sh E2E_C23_O2P e2e_c23_oracle e2e_c23_apg eqboc23p jsonb_tables_only '2021-04-01' 2021-05-01
```

```
nohup ./qbo_commandline_all.sh PRF_C4_O2P c4ora c4aur pqbopc04p jsonb_tables_only 5 &
```

Notes:

- Whenever we refresh Report template to add new tables added to the database, we need to refresh the following listfiles-
 - qbo_tables.txt (Tables names in upper case)
 - qbodata_tables.txt (Tables names in upper case)
 - qbodata_tables_jsonb.txt (7 QBO_DATA tables with JSONB column) (Tables names in upper case)
- This command takes care of uploading the generated Reports to S3. Please update the S3 bucket name if its not the same as [s3://bits-sbg-qbo-ppd-us-west-2/binaries/goldengate/dbsolo](https://s3.amazonaws.com/sbg-qbo-ppd-us-west-2/binaries/goldengate/dbsolo)
- sends consolidated Report in email
- sends consolidated Report data to Splunk.
- Allows to add or remove tables from the Report through above mentioned listfiles.
- Allows Template 3 to read comparison start date from an input file - **jsonb_start_end.txt** and run the report for last 45 days from the start date if Start & End dates have not been provided as input parameters
- Takes input parameter **<oos rerun counter (optional, default 3)>** to rerun oos report n number of times

12. Check the Logs and Reports in AWS S3 -

[s3://bits-sbg-qbo-ppd-us-west-2/binaries/goldengate/dbsolo](https://s3.amazonaws.com/sbg-qbo-ppd-us-west-2/binaries/goldengate/dbsolo)

How to update/refresh Report Template

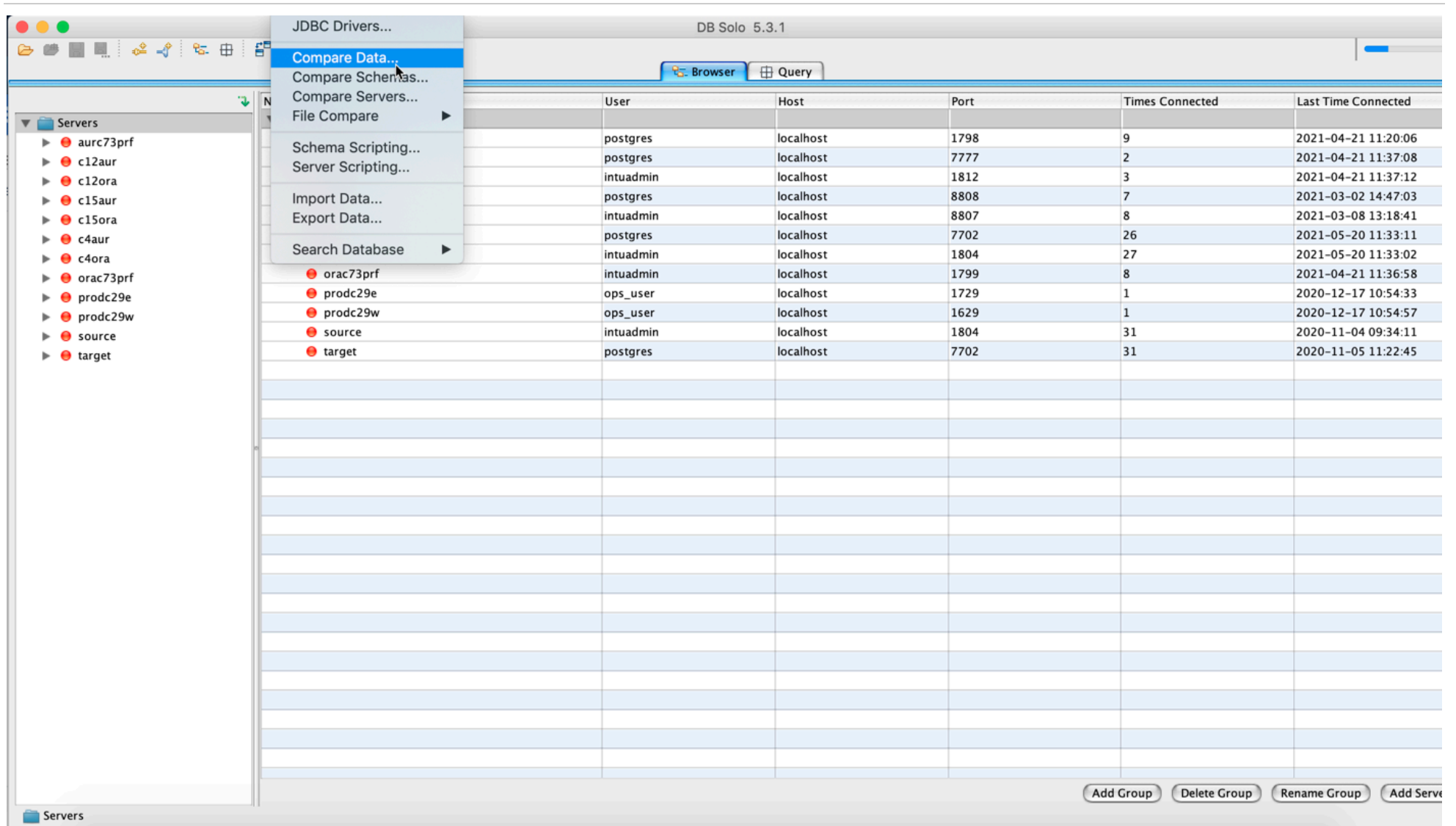
1. Copy the required xml project(s) from the git repo to your local mac or windows where you will use the DBSolo client

https://github.intuit.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/qbo_schema_tables_1.xml

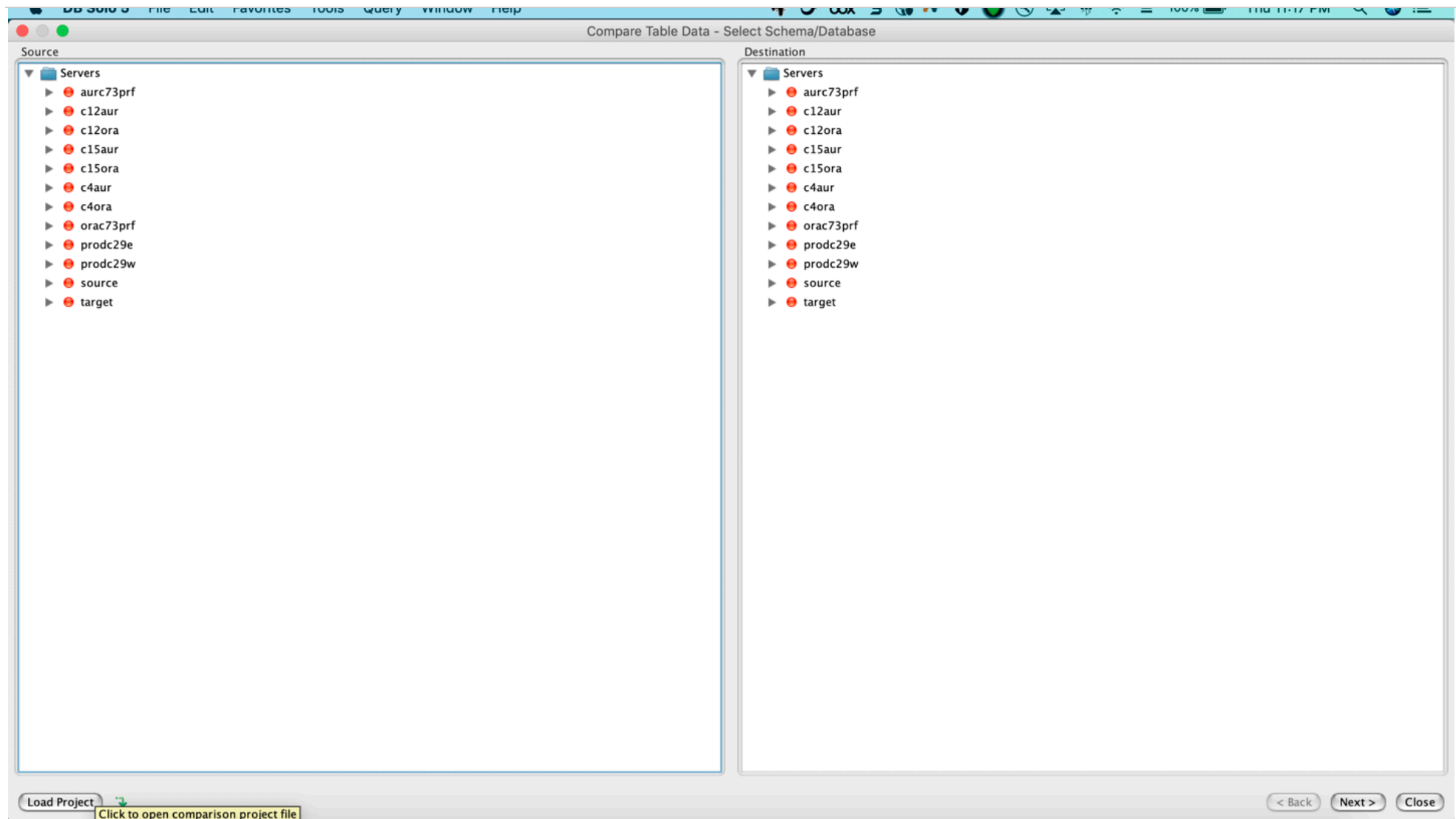
https://github.intuit.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/qbodata_schema_tables_1.xml

Notes: Please ensure it is the latest and no changes directly on the DBSolo ec2 instance

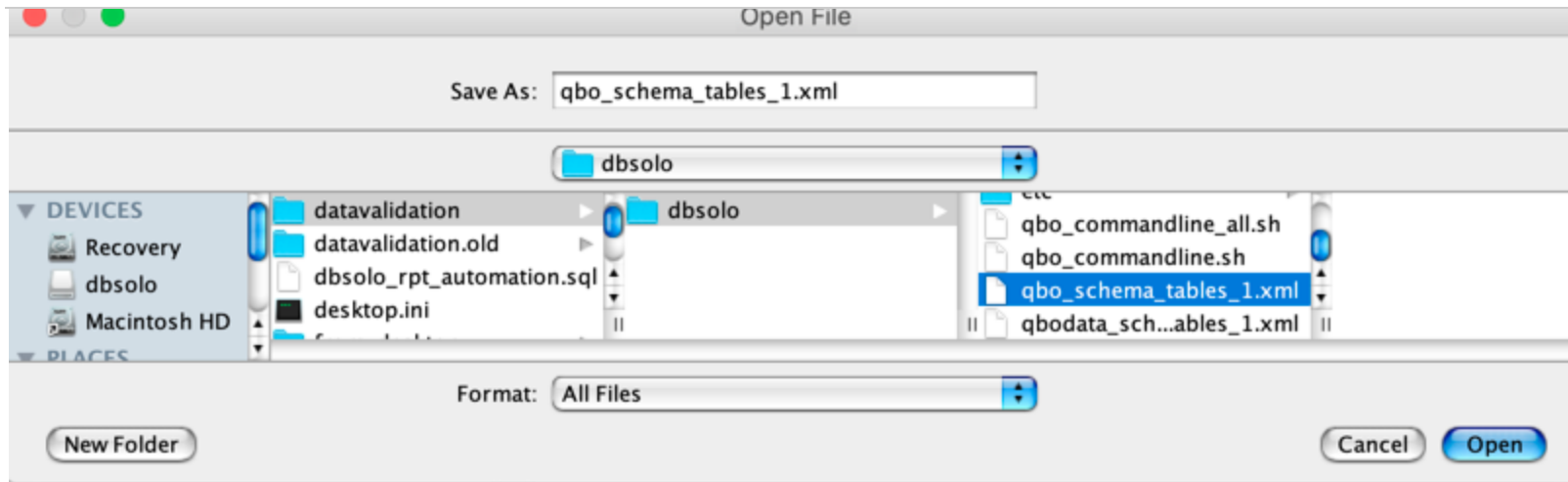
2. Open the DBSolo client and go to **Tools > Compare Data...**



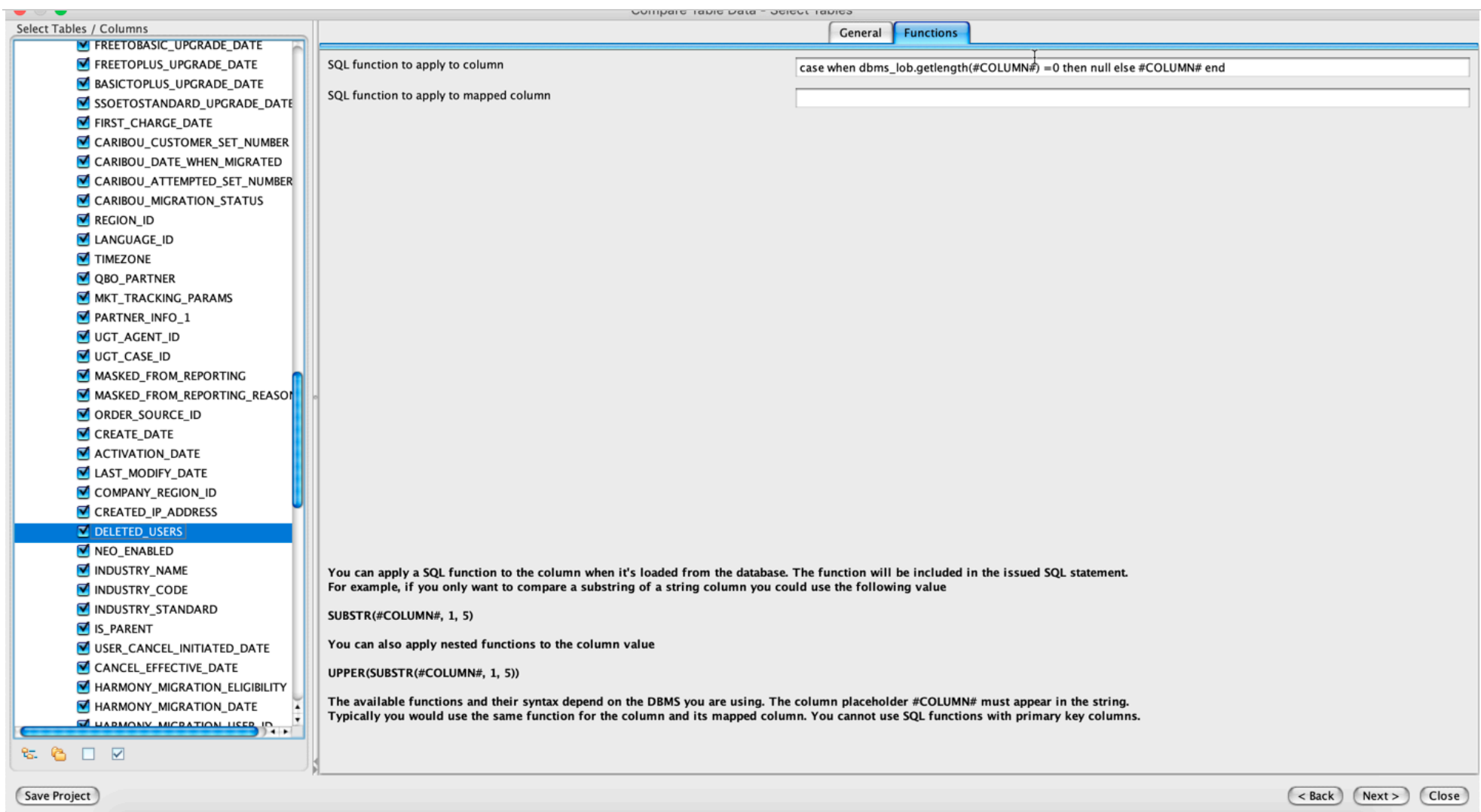
3. Click on the **Load Project** on the bottom left corner



4. Select the required xml project



5. Make the required change carefully (adding/removing tables/columns; adding/removing where clause; using/removing functions on top of the column etc) and then click on the **Save Project** on the bottom left corner.



6. Hit the **Save** on the next page unless you have to change any other setting.

Save Data Comparison Project

General HTML Options Advanced

Project File

/Users/psingh/Desktop/datavalidation/dbsolo/qbo_schema_tables_1.xml

Enter the name of the project file. You can also use the saved project file to run table data comparisons from command line.
Enter the following on command line to run the data comparison: `commandLine -dataCompare <projectfile>`

Synchronization Direction

☐ Create script to synchronize source side to be identical with destination side

☒ Create script to synchronize destination side to be identical with source side

Synchronization SQL Script

File Encoding UTF-8

☐ Single File

☒ File per table/view

File Name

Directory

Enter the name of data synchronization SQL file or directory. The command line tool will output the synchronization SQL to the specified file or directory.

SQL*Loader

Use SQL*Loader scripts to INSERT and UPDATE data ☐ Yes ☒ No

Directory for SQL*Loader files

Save Close

7. Compare this updated xml with the xml in git repo. Please ensure that you see the intended changes only and that, old settings and intact.

6. If a new table is added or if any change in the WHERE clause then you will have to regenerate/update the following scripts -

a.) If the Table has LAST_MODIFY_DATE or LAST_MODIFIED_UTC column for incremental comparison, please execute the below sql script on any QBO database with current schema version. **Replace the content of the below files with the sql resultset**

[03a_dbsolo_where_clause_generic_days.sh](#)

[03b_dbsolo_where_clause_generic_days_oos.sh](#)

[03c_dbsolo_where_clause_generic_date.sh](#)

[03d_dbsolo_where_clause_generic_date_oos.sh](#)

```
<sqlplus>
for 03a_dbsolo_where_clause_generic_days.sh refresh-
https://github.intuit.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/etc/generat

for 03b_dbsolo_where_clause_generic_days_oos.sh refresh-
https://github.intuit.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/etc/generat

for 03c_dbsolo_where_clause_generic_date.sh refresh-
https://github.intuit.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/etc/generat

for 03d_dbsolo_where_clause_generic_date_oos.sh refresh-
https://github.intuit.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/etc/generat
```

b.) If the Table has a date column (not in LAST_MODIFY_DATE or LAST_MODIFIED_UTC) for incremental comparison, update the below scripts manually.

[02a_dbsolo_where_clause_generic_days.sh](#)

[02b_dbsolo_where_clause_generic_days_oos.sh](#)

[02c_dbsolo_where_clause_generic_date.sh](#)

[02d_dbsolo_where_clause_generic_date_oos.sh](#)

c.) If a new table has been added or a table has been removed then you will have to update [the following listfiles by using the sqls mentioned below](#)

qbodata_tables_jsonb.txt (17 QBO_DATA tables with JSONB column)

```
For qbo_tables.txt & qbodata_tables.txt
<sqlplus>
https://github.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/etc/generat

For qbodata_tables_jsonb.txt
<sqlplus>
https://github.com/QBODataLayer/aurora-migration-prep/blob/master/database/datavalidation/dbsolo/etc/generat
```

Checkin these regenerated/updated files and, copy them to the DBSolo ec2 instance.

9. For Template 3 make sure jsonb_start_end.txt has the required start time for the comparisn.

10. Repeat **Step 2** to **Step 7**.

11. Do a test run

Other DBSolo WIKIs

[DBSolo QBO/QBO_DATA Tables Comparison Report & Details](#)

[Data Comparison between Oracle and Postgres Using DB Solo](#)

No labels

